



Final round

3

Revised 37-page NCLEX Study Guide

1. ABCs (Airway, Breathing, Circulation)
2. When in distress **DO NOT ASSESS!** Unless 2nd hand information is received.
3. Scenario
 - Expected outcome with Disease Process
 - Continue to monitor
 - Document finding
 - Unexpected finding with Disease Process
 - Nursing intervention that must make a difference
 - Call HCP
4. Mini Maslow's
 - 1) ABCs (& Pain unrelieved by meds)
 - 2) Safety
 - 3) Comfort (Pain)
 - 4) Psychological
 - 5) Social
 - 6) Spiritual
5. STAT words → Pick the answer that failing to do so will kill or cause great harm
 - Highest Priority
 - Most Important
 - Immediate Action
6. *Least Invasive First*
7. Secondhand Info → Any time you have 2nd hand info, the right answer is assess
 - UAP · Family
 - Labs · EMR
 - EKG · BP machine
8. Never ever take away the coping mechanism a patient uses during a crisis, **except** if the mechanism puts the patient or others @ risk
9. Eliminate answer choices & DNR
10. Never withhold Tx! If you're left with two answer choices and the options are to Tx, or watch the patient, Pick Tx!
11. Anytime there's a reversal from the norm, you must **worry!**
Ex: rebound tenderness (pain after you relieve pressure)
12. Stable Patients
 - UAP · LPN · New nurse
 - Graduate Nurse · Float Nurse · Travel nurse
13. Anytime you see excessive findings, That's not normal!
14. Always empower your patient
15. If a question has " ", pick an answer that has what they're feeling & not what they're saying
16. 3 R's of Psych
 - 1) Reality – Functional psych patient
 - 2) Reassure – Delirium
 - 3) Redirect – Dementia

Default Answers

1. Give meds either 1 hour before meal or 2 hours after meal
2. Give antacids 1 hour before med or 4 hours after med
3. When in doubt pick K (potassium)
4. 2 – 3 L of fluids
5. When in doubt pick answer that has you stay with patient
6. Anytime you see restless & ↓ level of consciousness = early sign always pick
7. Head of Bed → 30-45 degrees for any neuro patient
8. Elderly with acute onset confusion → UTI
9. Secretions will turn Orange/Red for meds
10. Anytime you have GI problem/exacerbation = NPO
11. All surgeries
 - 1st 24 hrs – bleeding
 - 48 hrs – infection
12. Check daily weights if it's a fluid problem
13. Lateral position for maternity
14. Remove answer choices that are 'absolutes'

Kaplan NCLEX Strategies

Kaplan RN Decision Tree

- Step 1 – Can you identify the topic of the question
- Step 2 – Are the answers assessment (get data) or implementation (to effect change)?
- Step 3 – Apply Maslow: Are the answers physical or psychosocial? (Physical trumps psychosocial)
- Step 4 – Are the answer choices related to ABCs?
- Step 5 – What is the outcome of each of the remaining answers?

Rules for Delegation

RN ASSIGNMENT

- Cannot delegate assessment, teaching, or nursing judgement

LPN/LVN ASSIGNMENT

- Assign stable with expected outcomes

UAP ASSIGNMENT

- Delegate standard, unchanging procedures

Five Rights of Delegation

RIGHT TASK – scope of practice, stable client

RIGHT CIRCUMSTANCES – workload

RIGHT PERSON – scope of practice

RIGHT COMMUNICATION – specific task to be performed, expected results, follow-up communication

RIGHT SUPERVISION – clear directions, intervene if necessary

Therapeutic Communication Tips

DO:	DO NOT:
• Do respond to feeling tone • Do provide information • Do focus on the client • Do use silence • Do use presence	• Do not ask 'why' questions • Do not ask 'yes/no' questions, except in the case of possible self-harm • Do not focus on the nurse • Do not explore • Do not say, "Don't worry!"

Who Do You See First?

Consider:

- Unstable vs. Stable
- Unexpected vs. Expected
- ABCs
- Acute vs. Chronic
- Actual vs. Potential

Common NCLEX Traps

- Do not ask “Why?”
- Do not leave the client.
- Do not persuade the client.
- Do not say, “Don’t worry!”
- Do not ‘do nothing.’
- Do not read into the question
- Do not pass the buck.

Strategies

- Only use textbook nursing – textbook knowledge
- Pain is psychosocial, unless, it’s severe, acute, & unrelenting
- If it’s a position question, is it going to prevent or promote something – **position, prevent, promote**
- Teaching/learning – use T/F on each answer
- Risk Questions – use Risk Factors
- If the answers have an absolute in them, do not pick them
- Question that have the phrase ‘And Then’ – did they miss something

Important Lab Values

WBC	4K – 11K
RBC	4 – 6
Hgb	12 – 16, Or 12-18
Hct	36 – 48, or 37-52
Plt	150K – 400K
BUN	8 – 20, or 7-22
Cr/Lithium	0.6 – 1.2
Urine Clearance	85 – 135, (GFR)=maintain above 60
Uric Acid	250 – 750 mg
Na	135 – 145
Cl	98 – 106
Ca	8.5 – 10.5
K	3.5 – 5.0
PO	2.5 – 4.5g
Mag	1.5 – 2.5, 4-7 if pregnant and receiving Toco

Warfarin	INR	2.0 – 3.5
Heparin	PT	10 – 13 Seconds
	PTT	25 – 35 Seconds

Therapeutic PTT: 1.5 – 2x the normal value (46 – 76 Seconds)

Cholesterol

HDL	x >	50
LDL	x <	100
Triglycerides	x <	150
Total Cholesterol	x <	200

Therapeutic Ranges

Dilantin Theophylline Acetaminophen	10 – 20
Digoxin	0.5 – 2.0

Albumin level 3.5 to 5.5

Acid-Base Balance

From the ass (diarrhea) –Metabolic Acidosis
From the mouth (vomit) –Metabolic Alkalosis

Potassium & Alkalosis – **ALKALOSIS**: K is LOW
– Acidosis is just the opposite: K is High

Arterial Blood Gases

1. Prior to drawing an ABG, perform the Allen's Test to check for sufficient blood flow
2. When drawing an ABG, the blood needs to be put in a heparinized tube.
 - Ensuring there are no bubbles.
3. Put on ice immediately after drawing, with a label.
 - The label should indicate if the pt was on room air, or how many liters of O2.

General Notes

- The person who hyperventilates is most likely to experience respiratory alkalosis.

Antidotes

- | | |
|--------------------------------------|--|
| • Aspirin | ☐ Activated Charcoal |
| • Coumadin (Warfarin) | ☐ Vitamin K |
| • Heparin | ☐ Protamine Sulfate |
| • Tylenol (Acetaminophen) | ☐ Mucomyst (acetylcysteine) – administered orally |
| • Digoxin (Lanoxin) | ☐ Digibind (immune Fab) |
| • Opioids | ☐ Narcan |
| • Iron overdose | ☐ Deferoxamine |
| • PCP | ☐ Activated charcoal |
| • Magnesium Sulfate | ☐ Calcium Gluconate |
| • TPA | ☐ Aminocaproic acid |
| • Pancuronium Br (NM blocking agent) | ☐ Neostigmine/Atropine |

Blood

For blood types:

- "O" is the universal donor (remember "o" in donor)
- "AB" is the universal recipient

Blood transfusion – sign of allergies in order:

- 1) Flank pain
- 2) Frequent swallowing
- 3) Rashes
- 4) Fever
- 5) Chills

Thrombocytopenia – Bleeding precautions!

- 1) Soft bristled toothbrush
- 2) No insertion of anything! (c/i suppositories, douche)
- 3) No IM meds as much as possible!

Sickle Cell Anemia

During **sickle cell crisis** there are two interventions to prioritize: **fluids and pain relief**.

Iron deficiency anemia – easily fatigued

1) **Fe PO (Iron)** - give with **Vitamin C** or on an **empty stomach**

2) **Fe via IM- Interferon via Z Track**

-- Peds: Kids are at risk for iron deficiency anemia if they ingest too much milk; >24oz/ day.

Pernicious Anemia - s/s include pallor, tachycardia, and **Sore Red, Beefy tongue**; will take Vit. B12 for life!

Shilling Test – test for pernicious anemia/ how well one absorbs Vit b12

General Notes

- A patient with a **low hemoglobin and/or hematocrit** should be evaluated for signs of **bleeding**, such as dark stools.

Burns

Rule of nines, 9 = head, 18 = arms, 36 = torso, 36 = legs, and 1 = perineum = 100%

The Parkland **formula** is a formula used for calculation the total **fluid** requirement in 24 hours for a burn patient
 $4\text{ml} \times \text{TBSA} \% (\text{Total Burn Surface Area}) \times \text{body weight (kg)} = \text{Total amount of fluid the patient will receive in 24 hrs}$

50% given in first eight hours

50% given in next 16 hours.

The **Number #1 Priority** for Burn Patients is maintaining a **patent airway**

1st Degree – Red and Painful

2nd Degree – **Blisters**

3rd Degree – No Pain because of blocked and burned nerves

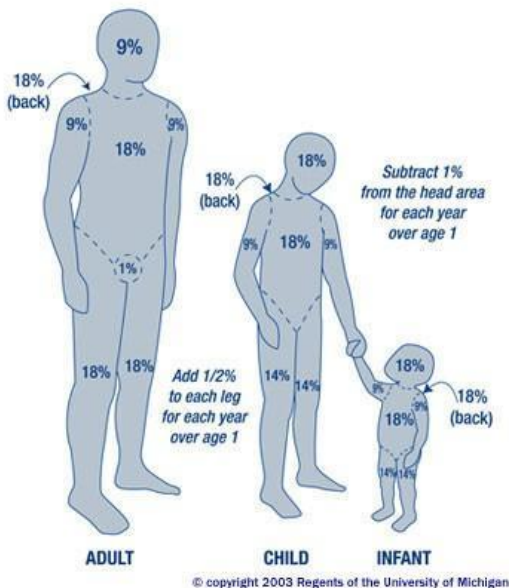
Cancer

A cancer patient is getting **radiation**. What should the nurse be most concerned about?

- Skin irritation? No.
- Infection kills cancer patients most because of the **leukopenia** caused by radiation.

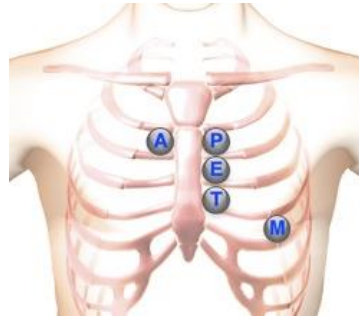
General Notes

- A breast cancer patient treated with **Tamoxifen** should report changes in visual acuity, because the adverse effect could be irreversible.
- Common sites for **metastasis** include the **liver**, brain, lung, bone, and lymph.
- **Bence Jones** protein in the urine confirms **multiple myeloma** (cancer of plasma cells)
- Patients with leukemia may have epistaxis (nosebleeds) b/c of low platelets



Cardiac

All	–	Aortic Valve
Physicians	–	Pulmonary Valve
Earn	–	Erb's Point
Their	–	Tricuspid Valve
Money	–	Mitral Valve (PMI)
Or APE To Man		



Cardiac Catheter

- Pre-Op – NPO 8-12hr prior, empty bladder, check pulses, tell pt they may **feel heat**, palpitations, or desire to cough with **dye injection**.
- Post Op – V/S, & **keep leg straight, bed rest 6-8 hrs, Sleep supine.**

General Notes

- **Blood tests** for MI: Myoglobin, CK and Troponin
- **Coarctation of the aorta** causes increased blood flow and **bounding pulses in the arms**
- Cor Pulmonale is right sided heart failure caused by left ventricular failure; (so pick **edema, JVD**, if it is a choice.)
- Normal **PCWP (pulmonary capillary wedge pressure)** is **8-13**. Readings of 18-20 are considered high.
- **Pulmonary sarcoidosis** (an inflammatory disease) leads to **right sided heart failure**.
- Anytime you see fluid retention. Think heart problems first.

Circulation

EleVate Veins; dAngle Arteries for better perfusion

For PVD remember DAVE (Legs are Dependent for Arterial & for Venous Elevated)

Virchow's Triad →		Risk Factors for DVT
V	–	Vascular Trauma
I	–	Increased Coagulability
R	–	Reduced Blood Flow

–Definitive diagnosis for **abdominal aortic aneurysm (AAA)** → **CT scan**

Fat Embolism S/S

- Blood tinged sputum (related to inflammation) <Pink frothy sputum>
- **increased erthyro sedimentation rate (ESR)**
- Respiratory alkalosis (related to tachypnea)
- **Hypocalcemia**, increased serum **lipids**
- "Snow Storm" effect on Chest x-ray

General Notes

- **Hypotension** and vasoconstriction meds may alter the accuracy of **O2 sats**.
- A **newly diagnosed hypertension** patient should have BP assessed in **both arms**

Cranial Nerves

Sensory=S Motor=M Both=B

1. Oh (Olfactory I)	Some
2. Oh (Optic II)	Say
3. Oh (Oculomotor III)	Marry
4. To (Trochlear IV)	Money
5. Touch (Trigeminal V)	But
6. And (Abducens VI)	My
7. Feel (Facial VII)	Brother
8. Very (Vestibulocochlear/Auditory VIII)	Says
9. Good (Glossopharyngeal IX)	Big
10. Velvet (Vagus X)	Brains
11. Such (Spinal Accessory XI)	Matter
12. Heaven (Hypoglossal XII)	More

On Old Olympus Towering Top A Finn And German Viewed Some Hopes

Cultural

Greek heritage - they put an amulet or any other use of protective charms around their baby's neck to avoid "evil eye" or envy of others

Lyme Disease is found mostly in Connecticut

Jewish Folks: no meat and milk together

Diabetes

Blood Sugar ~ Hyperglycemia – Hot & Dry ~ Sugar High
Hypoglycemia – Cold & Clammy ~ Need some candy

To remember how to draw up INSULIN think: **Nicole Richie RN** <Regular is clear & don't wanna put dirty needle in clear so Regular<CLOUDY> is pulled in first>
Air into **NPH**, then air into **Regular**, draw up **Regular insulin** then draw up **NPH**

Oral Hypoglycemics

- Do not attempt to give an oral hypoglycemic to an unconscious pt, as this poses the risk of aspirations
- A typical adverse reaction is **rash**, photosensitivity.

HbA1c – test to assess how well blood sugars have been controlled over the past 90-120 days.

4- 6 corresponds to a blood sugar of 70-110;

7 is ideal for a diabetic and corresponds to a blood sugar of 130

Fluids are the most important intervention with **HHNS** as well as **DKA**, so get fluids going first.

DKA

- While treating **DKA**, bringing the glucose down **too far and too fast** can result in increased **intracranial pressure** due to water being pulled into the CSF.
- **Serum acetone and serum ketones** rise in **DKA**.
- As you treat the acidosis and dehydration expect the **potassium to drop** rapidly, so be ready, with potassium replacement.

HHNS

- With HHNS there is no ketosis, and no acidosis.
- Potassium is low in HHNS (d/t diuresis)

General Notes

- Extra insulin may be needed for a patient taking **Prednisone** (remember, steroids cause increased glucose).
- **Second voided urine most accurate when testing for ketones and glucose.**

Drugs

General Notes

- Give NSAIDS, Corticosteroids, drugs for **Bipolar**, Cephalosporins, and Sulfonamides **WITH food.**
- Best time to take Growth Hormone PM (Octreotide), Steroids AM, Diuretics AM, Aricept (Donepezil) AM - for Alzheimer's disease.
- **Antacids are given after meals**
- Remember the action of **vasopressin** because it sounds like "press in", or **vasoconstrict**.
- **If mixing antipsychotics (i.e. Haldol, Thorazine, Prolixin) with fluids, meds are incompatible with caffeine and apple juice**
- The main **hypersensitivity** reaction seen with **antiplatelet** drugs is **bronchospasm** (anaphylaxis) - "think NSAIDS causing bronchoconstriction in asthma patients"
- **Glucagon** increases the effects of oral **anticoagulants**.
- All psych meds' (except Lithium) side effects are the same as SNS but the BP is decreased
 - **SNS- Increase in BP, HR and RR** (dilated bronchioles), dilated pupils (blurred vision), Decreased GUT (urinary retention), GIT (constipation), Constricted blood vessels and Dry mouth.

Anti-Anemics

- Iron injections should be given Z-track, so they don't leak into SQ tissues.
- Take iron elixir with juice or water.... **never with milk**

Antiarrhythmics

- **Verapamil**: a **calcium channel blocker**, used to treat hypertension, angina; **assess for constipation**
- **Digoxin**: Check pulse, if it's less than 60, hold medication, prior to administration check both potassium and dig levels
 - Pick '**do vitals**' before administering that **dig**. (apical pulse for one full minute).
 - Making sure that patients on **Digoxin and Lasix** are getting **enough potassium**, because low potassium potentiates Digoxin toxicity and can cause dysrhythmias.
 - **Digitalis increases ventricular irritability** and could convert a rhythm to v-fib following cardioversion.
- **Adenosine**: is the treatment of choice for paroxysmal **atrial tachycardia**.
- **Flecainide (Tambocor)**: *Antiarrhythmics* med, **limit fluids** and sodium intake, because sodium increases water retention which could lead to heart failure.

Antianxiety

- **Diazepam** is a commonly used tranquilizer given to reduce anxiety before OR
- **Midazolam**: an anesthetic given for conscious sedation, **watch out for respiratory depression and hypotension**
- **Chlordiazepoxide**: **treatment of alcohol withdrawal**; don't take alcohol with this medication, causes nausea & vomiting
- **Hydroxyzine**: **treatment of anxiety as well as itching**, commonly administered pre-op, **watch out for dry mouth**
- **Lorazepam**: **treatment of choice for status epilepticus**

Anti-asthmatics

- **INtal<cromolyn sodium aerosol>**: an inhaler used to treat allergy induced asthma may cause bronchospasm, think... **IN**to the asthmatic lung

Antibiotic

- **Aminoglycosides**: Adverse Effects are bean shaped - Nephrotoxic to Kidneys and Ototoxic to Ears
 - __Mycin (drugs that end in or have in their name); except erythromycin (have -thro- in drug name)
- **Sulfamethoxazole/trimethoprim**: an antibiotic; **common side effect is diarrhea (drink plenty of fluids)**; **do not take if allergic to sulfa drugs**

Anticholinergic---- INE ending**

- Anticholinergic effects — assessment
 - dry mouth – can't spit
 - urinary retention – can't piss
 - constipated – can't shit
 - blurred vision – can't see
- **Atropine used to decrease secretions** & asystole
 - **Atropine blocks acetylcholine** (remember acetylcholine reduces secretions)
 - Atropine OD
 - Hot as a Hare (Temp)
 - Mad as a Hatter (LOC)
 - Red as a Beet (flushed face)
 - Dry as a Bone (Thirsty)
- **Benztropine: Treats Parkinson as well as extrapyramidal side effects of other drugs**

Anticonvulsants

- **Phenytoin <Dilantin>**: treatment of seizures; therapeutic drug level: **10-20**; **contraindicated during pregnancy**; Side effects include rash (stop med if seen), **gingival hyperplasia (can be prevented w/ good hygiene)**.
 - **Dilantin Toxicity** ☐ **poor gait + coordination, slurred speech, nausea, lethargy, & diplopia**
- **Phenobarbital**: **treatment of epilepsy**; can be taken **during pregnancy**

Antidepressants

- **Zoloft/Sertraline**: side effects include agitation, sleep disturb, and dry mouth (**SSRI**)
- MAOI's: antidepressant
 - **An easy way to remember MAOI'S! think of PANAMA!**
 - PA – parnate- Tranylcypromine**
 - NA – nardil- Phenelzine**
 - MA – marplan- Isocarboxazid**
 - MAOI's used for depression all have an arrr sound in the middle (Parnate, Marplan, Nardil) – Remember that Pirates say arrr, so think “pirates take MAOI's when they're depressed”
 - They have metallic bitter taste

Antifungal

- **Amphotericin B**: This medication **causes hypokalemia** (amongst many other side effects as well); patient will most likely get a fever; **pre-medicate with acetaminophen and/or diphenhydramine (preferably both) before administering to a patient**

Antiemetic should

- **Trimethobenzamide <Tigan>**: Treatment of postop nausea and vomiting, and for nausea associated with gastroenteritis
- **Promethazine <Phenergan>**: an antiemetic used to reduce nausea

Anti-gout Agents

- **Probenecid, Colchicine, Allopurinol**
- **Allopurinol**: Push with fluids, in order to flush the uric acid out of system; **DO NOT TAKE W/ VITAMIN C**

Antihypertensives

- **hydralazine**: treatment of HTN or CHF, Report flu-like **symptoms**, rise slowly from sitting/lying position; **take with meals.**

Antimanic

- **Lithium**:
 - L-level of therapeutic effect is 0.5-1.5*
 - I-indicate mania*
 - T-toxic level is 2-3** - nausea & vomiting, diarrhea, tremors
 - H-hydrate 2-3L of water/day*
 - I-increased Urinary output and dry mouth*
 - U-uh oh; **give Mannitol and Diamox if toxic signs and symptoms are present***
 - M-maintain Na intake of 2-3g/day**

Antimetabolites

- **Hydroxyurea**: treatment of sickle cell & certain types of leukemia; when used to Tx sickle cell, report GI **symptoms** immediately, could be sign of toxicity

Antineoplastic

- **vincristine**: treatment of leukemia; given IV ONLY
- **Asparaginase**: treatment for acute lymphoblastic leukemia; Test for hypersensitivity prior to administration

Antiparkinsonian Agents

- **Carbidopa-Levodopa**: treatment of Parkinson; side effects include drowsiness and the patient's sweat, saliva, **urine may occasionally turn reddish brown; contraindicated with MAOI's**
- **Trihexyphenidyl** treatment of Parkinson, **causes sedation**
- **Levodopa**: **Contraindicated in patients' w/ glaucoma, avoid B6**

Antipsychotics

- **Risperidone**: **Doses over 6mg** can cause **tardive dyskinesia**, this is a first line antipsychotic in children
- **Clozapine**: Side effects include **agranulocytosis**, tachycardia, and seizures, **WATCH FOR INFECTION***
- **Thiothixene**: treatment of schizophrenia; **assess for EPS**
- **Haloperidol**: preferred antipsychotic in **elderly**, but it has a high risk of extrapyramidal side effects (dystonia, tardive dyskinesia, tightening of jaw, stiff neck, swollen tongue, **later on swollen airway**)
 - The nurse must monitor for early signs of reaction and **give IM Benadryl**
- **1st generation antipsychotics are the leading cause of Akathisia**
 - **Akathisia** is characterized by motor restlessness, i.e. a need to keep going
 - Can be mistaken for agitation
 - Treated with Anti Parkinson's meds

Anti-rheumatics

- **Indomethacin**: an NSAID; treatment of **arthritis** (osteo, rheumatoid, gouty), bursitis, and tendonitis.

Antispasmodics

- **dicyclomine**: treatment of irritable bowel; assess for anticholinergic side effects.

Antitubercular

- **Rifampin**: Red orange tears and urine (b/c it dyes bodily fluid orange); contraceptives don't work as well
- **Ethambutol**: messes with your Eyes
- **Isoniazid (INH)**: treatment & prevent TB; it can cause **peripheral neuritis/neuropathy** (nerve damage); do not give with Phenytoin □ **can cause phenytoin toxicity; monitor LFT's; give B6 along with**; hypotension will occur initially, then resolve
- **TB drugs are liver toxic (hepatotoxic)**.
 - An adverse reaction is **peripheral neuropathy**
 - Ask patients if they have Hep B

Antithyroid

- **PTU and Tapazole**: Tx of **hyperthyroidism** & prevention of thyroid storm
- **Lugol's Solution**: adjunct Tx for **hyperthyroidism** as well as radiation protectant. **An adverse reaction: Burning sensation** in the mouth, and brassy taste. Report it to the doctor.

Antiulcer

- **Aluminum hydroxide**: treatment of GERD and kidney stones, watch out for constipation.
 - Long term use of **amphogel** (**binds to phosphates, increases Ca, robs the bones...leads to increased Ca reabsorption from bones** □ WEAK BONES)
 - **Amphogel and Renegal should be taken with meals**
- **Sucralfate**: treatment of duodenal ulcers, this medication coats the ulcer by creating a mucosal barrier, so the patient should take this medication before meals; be aware of **constipation** as a potential side effect
- **Cimetidine**: an H2 antagonist **taken with food**; use cautiously in the elderly population; interacts with a lot of other drugs
- **Peptic ulcers caused by H. pylori are treated with Flagyl, Prilosec and Biaxin**. This treatment kills bacteria and stops production of stomach acid but does not heal ulcer.

Antiviral

- **Ganciclovir**: used for retinitis caused by cytomegalovirus, patient will need regular eye exams, report dizziness, confusion, or seizures immediately

Anthelmintic/Anti-worm

- **Mebendazole**: Administer this medication with a high fat diet as this increase's absorption

Beta Blockers

- **Timolol**: treatment of glaucoma

Bronchodilators

- **Theophylline**: Tx of **asthma or COPD**; **therapeutic drug level is 10-20**; increases the risk of digoxin toxicity and decreases the effects of lithium and Phenytoin; causes **GI upset**, give with food

CNS Stimulants

- **Dexedrine**: treatment of **ADHD**; may alter insulin needs; avoid taking with MAOI's; **take in morning** (insomnia possible side effect)
- **Methylphenidate**: Tx of **ADHD**; **assess for heart related side effects**, report them immediately; **child may need a drug holiday b/c it stunts growth**.

Digestive Agent

- **Pancrealipase**: These are **pancreatic enzymes**, which are to be taken with each meal! Not before, not after, but **W/** each meal.

Diuretics

- **Mannitol (osmotic diuretic)**: used for **Head injuries**; it crystallizes at room temp so **ALWAYS** use filter needle
- **Acetazolamide**: Tx of **glaucoma**, & high-altitude sickness; do not take if allergic to sulfa drugs; may cause hypokalemia
- **Lasix**: Tx of edema due to heart failure; can cause a patient to lose his appetite (**anorexia**) due to reduced potassium
 - Patients receiving **Lasix and Dig**, need to be getting **enough potassium**, b/c low potassium potentiates Dig Toxicity and can cause dysrhythmias.

Hormones

- **Levothyroxine**: Tx of **hypothyroidism**, this medication may take several weeks to take effect; notify doctor of chest pain; **take in the AM on empty stomach**; may cause hyperthyroidism
 - **Insomnia** is a side effect of thyroid hormones (Ex: Synthroid) □ Increases met. rate, your body is "too busy to sleep" as opposed to the folks with hypothyroidism who may report somnolence (dec. met rate, body is slow and sleepy).
- **Conjugated estrogens**: treatment occurs after menopause for estrogen replacement

Hypokalemic

- **Sodium Polystyrene Sulfonate**: When giving administering this drug, we need to worry about dehydration (K has inverse relationship with Na)
 - **Don't use this medication if patient has hypoactive bowel sounds.**

Opioid Analgesics

- **Meperidine <Demerol>**: Tx for moderate to severe pain; used for patients with pancreatitis (these patients could NOT receive morphine sulfate); **Do not give Demerol to pts. with sickle cell crisis**

Pediculocides

- **Lindane <Antiparasite>**: Tx of **scabies and lice**;
 - **Scabies ~ apply lotion once and leave on for 8-12 hours**
 - **Lice ~ use the shampoo and leave on for 4 minutes with hair uncovered then rinse with warm water and comb with a fine-tooth comb**

Skeletal Muscle Relaxant

- **Dantrolene**: treatment for spasticity, associated w/ spinal cord injury, stroke, cerebral palsy, or MS; it may **take a week** or more to be effective.
- **succinylcholine**: is used for **short-term neuromuscular** blocking agent for procedures like intubation and ECT.
- **Vecuronium Bromide**: an adjunct medication for general anesthesia-induced muscle relaxation for endotracheal intubation/mechanical ventilation, it's used for intermediate or long-term.

Statins

- **Simvastatin's**: **Treatment for hyperlipidemia**; **take on empty stomach** to enhance absorption; report any unexplained muscle pain, especially if fever is present

Sympathomimetic

- **Dopamine (Intropine)**: Tx of hypotension, shock, low cardiac output, poor perfusion to vital organs; monitor EKG for arrhythmias, and BP

Vasodilators

- **Nitroprusside**: When the patient is on this medication, **monitor thiocyanate (cyanide)**. The **normal value should be 1, >1 is heading toward toxicity**

Endocrine Diseases

Addison's: **hypoNa, hyperK**, hypoglycemia, **dark pigmentation**, decreased resistance to stress, fractures, alopecia, **weight loss**, GI distress – **Addison's disease** (need to **"add"** hormone)

Blood pressure is the most important assessment parameter in Addison's, as it causes severe **hypotension**.

Addisonian Crisis Nausea & vomiting, confusion, abdominal pain, extreme weakness, hypoglycemia, dehydration, decreased BP

Managing stress in a patient with adrenal insufficiency (**Addison's**) is paramount, because if the adrenal glands are stressed further it could result in Addisonian crisis.

Cushing's: **hyperNatremia, hypoKalemia**, hyperglycemia, prone to infection, muscle wasting, weakness, edema, HTN, hirsutism, moon-face/buffalo hump – **Cushing's syndrome** (have extra **"cushion"** of hormones)

	Sex	Salt	Sugar
Addison's	↓	↓	↓
Cushing's	↑	↑	↑

OR

Addison's= down, down, down, up, down

Cushing's= up, up, up, down, up

Addison's= **hyponatremia**, hypotension, decreased blood vol, hyperkalemia, hypoglycemia

Cushing's= **hypernatremia**, hypertension, increased blood vol, hypokalemia, hyperglycemia

Addison's disease (need to **"add"** hormone)

Cushing's syndrome (have extra **"cushion"** of hormones)

Diabetes Insipidus (decreased ADH): excessive urine output and thirst, dehydration, weakness; **administer Vasopressin**
SIADH (increased ADH): change in LOC, decreased deep tendon reflexes, tachycardia, n/v/a, HA; **administer Declomycin, diuretics**

Hyper-parathyroid: fatigue, muscle weakness, renal calculi, back and **joint pain (increased calcium)**; **diet should consist of low Ca, & high phosphorus diet (Calcium and phosphorus has inverse relationship)**

- **Polyuria** is common with the **hypercalcemia** caused by hyperparathyroidism.

Hypo-parathyroid: CATS – convulsions, arrhythmias, tetany, spasms, stridor, & **decreased calcium**; diet should consist of high Ca, & low phosphorus diet

Hyperthyroidism/Graves' disease: accelerated physical and mental function; sensitivity to heat, fine/soft hair

For **HYPER**thyroidism think of **MICHAEL JACKSON in THRILLER! SKINNY, NERVOUS, BULDGING EYES**, up all night, heart beating fast

Thyroid Storm: is **HOT** (hyperthermia), **↑ HR**, & **HTN**

Hypothyroidism/Myxedema: slowed physical and mental function, sensitivity to cold, hypothermia, dry skin and hair

Post-thyroidectomy: Must watch for **hypercortisolism** and temporary diabetes insipidus. **Position the patient in semi-Fowler's, prevent neck flexion/hyperextension, and have trach at bedside**

Pheochromocytoma: hypersecretion of too much of epi/norepi, persistent HTN, increased HR, hyperglycemia, diaphoresis, tremor, pounding heart; avoid stress, frequent bathing and rest breaks, avoid cold and stimulating foods, surgery to remove tumor

Pancreatitis: Pt is placed in fetal position, maintain NPO, gut rest, prepare antecubital site for PICC b/c will probably be receiving TPN/Lipids. After pain relief, cough and deep breathe is important because of fluid pushing up in the diaphragm.

Hepatitis

Hepatitis A = –ends in a **VOWEL**, comes from the **BOWEL** (Hep A)

Hepatitis B = Blood and Bodily fluids

- Anaphylactic reaction to baker's yeast is contraindication for Hep B vaccine.

Hepatitis C = is just like B

- During the acute stage of **Hep-A gown and gloves** are required.
- In the convalescent stage it is no longer contagious.

Eyes & Ears

Ears

- Pull pinna **down and back** for kids < 3 yrs. when instilling eardrops

Meniere's Disease

- Tx: Admin diuretics to decrease endolymph in the cochlea
- Nursing Care: restrict Na, lay on affected ear when in bed

Triad:

- 1)Vertigo
- 2)Tinnitus
- 3)Nausea & vomiting

Strabismus Treatment is BOTOX

- Patch the **GOOD eye**, so that the weaker eye can get stronger.
- Botox can be used with strabismus to relax vocal cords in spasmodic dysphonia.

Eyes

OU – Both eyes

OS – Left eye

OD – Right eye (**dominant Right eye** – just a tip to remember)

General Notes for Eyes

- Assessing extraocular eye movements check cranial nerves **3 (oculomotor), 4 (trochlear), and 6 (Abducens)**.
- Glaucoma intraocular pressure is greater than the **normal (22 mm Hg)**, give miotic to constrict (pilocarpine), **NO ATROPINE**
- Apply eye drop to conjunctival sac and afterwards apply pressure to nasolacrimal duct / inner canthus

Fluid & Electrolyte Imbalances

Hypovolemia (FVD) – increased temp, rapid/weak pulse, increase **respiration**, hypotension, anxiety, urine specific gravity >1.030. (More U Specificity means urine more concentrated. More particles in urine and less dilution)

Hypervolemia (FVE) – bounding pulse, SOB, dyspnea, rales/crackles, peripheral **edema**, hypertension, urine specific gravity <1.010; Semi-Fowler's

Fluid volume overload caused by IVC fluids infusing too quickly and CHF can cause an **S3 heart sound**.

Hyponatremia: nausea, muscle cramps, increased ICP, muscular twitching, convulsion; osmotic diuretics, fluids
Hypernatremia (greater than 145): increased temp, weakness, disorientation/delusions, hypotension, tachycardia; hypotonic solution

Remember....

Hypernatremia presents similarly to **Fluid Volume Deficit**

Hyponatremia presents similarly to **Fluid Volume Excess**

Remember SALT...

Skin flushed

Agitation

Low grade fever

Thirst

Hypokalemia: muscle weakness, dysrhythmias, increase K (raisins, bananas, apricots, oranges, beans, potatoes, carrots, celery)

No Pee, no K (do not give **potassium** without adequate urine output)

Hyperkalemia: MURDER – Muscle weakness, Urine (oliguria/anuria), Respiratory depression, Decreased cardiac contractility, ECG changes, Reflexes

Hypocalcemia <Non sedation>: CATS – convulsions, arrhythmias, tetany, spasms and stridor

Hypercalcemia <Sedative effect>: muscle weakness, lack of coordination, abdominal pain, confusion, absent tendon reflexes, sedative effect on CNS

HypoMg <Non sedative effect>: tremors, tetany, seizures, dysrhythmias, depression, **confusion, dysphagia**; dig toxicity (**not enough magnesium, everything goes up**)

HyperMg <Sedative effect>: depresses the CNS, hypotension, facial flushing, **muscle weakness, absent deep tendon reflexes**, shallow respirations, emergency (**More magnesium, everything goes down**)

Fundamental Skills

Order of Assessment

Inspection

Auscultation

Palpation

Percussion

Order of Assessment for Abdomen/Children

Inspection

Auscultation

Percussion

Palpation

· Bowels Sounds may be obstructed and not heard, if performed out of order

· In Kids, go from least to most invasive

If your normally lucid patient starts **seeing bugs** you better check his **respiratory** status first. The first sign of hypoxia is **restlessness**, followed by agitation, and things go downhill from there all the way to delirium, hallucinations, and coma. **So, check the o2 stat, and get ABG's if possible.**

The immediate intervention after a **sucking stab wound** is to dress the wound and tape it on three sides which allows air to **escape**. Do not use an occlusive dressing, which could convert the wound from open pneumo to closed one, and a tension pneumothorax is worse situation. **After that get your chest tube tray, labs, iv.**

An example of when you would implement before going through a bunch of assessments is when someone is experiencing **anaphylaxis**. Get the **ordered epinephrine** in them stat, especially if the stem clearly states the s/s (difficulty breathing, increasing anxiety, etc.)

Radioactive iodine – The key word here is **flush**. **Flush substance out of body w/3-4 liters/day for 2 days and flush the toilet twice after using for 2 days.** Limit contact w/patient to 30 minutes/day. **No pregnant visitors/nurses, and no kids.**

Role-Relationship Pattern, a nursing diagnosis focused on the person's roles in the world and relationships with others. **To access the role relationship pattern, focus on image, and relationships with others.**

Bleeding is part of the '**circulation**' assessment of the **ABCD's** in an emergent situation.

- Therefore, if airway and breathing are accounted for, a compound fracture requires assessment before Glasgow Coma Scale and a neuro check (D=disability, or neuro check)

Potassium

- The vital sign you should check first with **high potassium** is **pulse** (due to dysrhythmias).
- **Never give potassium** if the patient is **oliguric** or anuric (because can't pee out the potassium = hyperkalemia)

NG Tube

- An **NG tube** can be irrigated with **cola** and should be taught to family when a client is going home with an NG tube.
- An **antacid** should be given to a mechanically ventilated patient with an NG tube if the pH of the **aspirate** is **<5.0** (because pH is low, acidic)
 - **Aspirate should be checked at least every 12 hrs.**

Hemovac

- Can be used after mastectomy
- How to Clean/Empty:
 - **Empty when full or q8hr**, remove plug, empty contents, place on flat surface, cleanse opening and plug with alcohol sponge, **compress evacuator completely to remove air**, release plug, check system for operation.

Liver

- Liver Biopsy:
 - **Prior to a liver biopsy it's important to be aware of the lab result for prothrombin time**
 - **NPO for 6 hrs** morning of biopsy, & administer vitamin k (for clotting factors), as well as a sedative
 - **Teach patient that he will be asked to hold breath for 5-10sec**, supine position, lateral with upper arms elevated.
 - **Post Op – position on right side**, frequent vital signs, report severe abdominal pain stat, **no heavy lifting 1 week**.
- A patient with liver **cirrhosis and edema** may **ambulate**, then sit with legs elevated to try to mobilize the edema.
- For esophageal varices, a Sengstaken Blakemore tube is used, **keep scissors at bedside (to cut the tube in an emergency situation) - U world question!!!**
- **Tylenol poisoning – liver failure possible for about 4 days**. Close observation required during this timeframe, as well as treatment with **Mucomist (Tylenol/acetaminophen antidote)**.

Paracentesis: (removing a ton of fluid from abdomen from liver failure)

- Pre-Op – The **patient should empty their bladder**
- Post Op – **Vital signs**, report elevated temperature (for infection), observe for signs of hypovolemia.

MRI

- Claustrophobia
- **No metal**
- **Assess pacemaker**

Laparoscopy - (fiber-optic instrument is inserted through the abdominal wall to view the organs in the abdomen or to permit a surgical procedure)

- **CO2 used to enhance visual**
- **General anesthesia** is administered, and a **foley** is inserted
- **Post Op – Walk w/ patient to decrease CO2 build up used for procedure.**

Compartment Syndrome ~ an EMERGENCY situation

- **Paresthesia and increased pain** are classic symptoms!!!! <No relief from Opioids>

- Neuromuscular damage is irreversible, 4-6 hours after onset

General Notes

- For patients with Halo device; Remember safety first & have a screwdriver nearby. (Keep the pins infection free)
- **Iatrogenic** means it was **caused** by treatment, procedure, or medication.
- A **3-way occlusive dressing** is used if a **chest tube** is accidentally pulled out of the patient.
- **Cultures are obtained before starting IV antibiotics!!!!** (what would you do first!?!)
- **Orthostatic hypertension** is verified by a **drop**-in pressure with **increasing** heart rate
- You will ask **every new admission** if he has an **advance directive**, and if not, you will explain it, and he will have the option to sign or not.
- **A guy loses his house in a fire. Priority is using community resources to find shelter, before assisting with feelings about the tremendous loss.** – (Maslow)
- **No nasotracheal suctioning with head injury or skull fracture** (increases intracranial pressure!!!)
- **Feed upright** to avoid otitis media.
- **Water intoxication** will be evidenced by **drowsiness**, and altered mental status, in patients with TURP syndrome, or as an adverse reaction to desmopressin (for diabetes insipidus).
- **Other than initially to test tolerance, G-tube and J-tube feedings are usually given as continuous feedings.**
- **Four side-rails up** can be considered a form of **restraint**. Even in LTC (long term care) facility when a client is a fall risk, keep lower rails down, and one side of bed against the wall, lowest position, wheels locked.

Gastrointestinal

Dumping syndrome: **increase fat and protein**, small frequent meals, lie down after meal to decrease peristalsis, wait 1 hr after meals to drink!!!! (know!!!)

Weighted NI (Naso intestinal tubes) must float from stomach to intestine. Don't tape the tube right away after placement, may leave coiled next to patient on head of bed. Position patient on RIGHT to facilitate movement through pylorus.

After **g-tube placement** the stomach contents are drained by gravity for **24 hours** before it can be used for feedings.

Stomach

- Mucus in ileal conduit is expected.
- **Dusky Stoma = Poor blood supply**
- **Protruding = Prolapsed**
- Sharp pain + Rigidity = Peritonitis

General Notes

- Don't fall for 'reestablishing a normal bowel pattern' as a **priority with small bowel obstruction**. Because the patient can't take in oral fluids, '**maintaining fluid balance**' comes first. "think ABC's!!"
- **Gastric Ulcer pain** occurs 30 minutes to 90 minutes after eating, not at night, and doesn't go away with food. **Duodenal ulcer** pain goes away with food.
- Cushing's ulcers related BRAIN injury & increased intracranial pressure.
- **When you see Coffee-brown emesis, think peptic ulcer.**
- **Patients should not have cantaloupe before an occult stool test; because cantaloupe is high in both vitamin C, which causes a false positive for occult blood!**

TABLE 38-2		
Glasgow Coma Scale		
BEHAVIOR	RESPONSE	SCORE
Eye opening response	Spontaneously	4
	To speech	3
	To pain	2
	No response	1
Best verbal response	Oriented to time, place, and person	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
	No response	1
Best motor response	Obeys commands	6
	Moves to localized pain	5
	Flexion withdrawal from pain	4
	Abnormal flexion (decorticate)	3
	Abnormal extension (decerebrate)	2
	No response	1
Total score:		
Best response		15
Comatose client		8 or less
Totally unresponsive		3

Glasgow Coma Scale – Eyes, Verbal, Motor

Remember... < than 8 intubate

It is similar to measuring dating skills...**max 15 points** -one can do it if **below 8** you are in **Coma**.

EYES: So, to start dating you got to **open** your **EYES first**, if you able to do that spontaneously and use them correctly to SEE whom you dating you earn 4. But if she has to scream on you to make you open them it is only 3....and 1 you don't care to open even if she tries to hurt you.

VERBAL: If you get good EYE contact (4 points) then move to **VERBAL**. Talk to her/ him! If you can do that You are really **ORIENTED** in situation she/he unconsciously gives you 4 points! if you like her try not to be **CONFUSED** (3), and of course do not use **INAPPROPRIATE WORDS** (3), she will not like it, try not to **RESPOND WITH INCOMPREHENSIBLE SOUNDS** (2), if you do not like her – just show no **VERBAL RESPONSE**(1)

MOTOR: Since you've got EYE and VERBAL contact you can **MOVE** now using your Motor Response Points. This is VERY important since Good moves give you 6!

Decorticate is toward the 'core'.

Decerebrate the other way (out).

Decorticate positioning in response to pain = **Cortex** involvement

Decerebrate in response to pain = Cerebellar, **brain stem** involvement (this is the worst one)

Hallmark Signs of Symptoms

01. **Pulmonary TB (pulm-TB)** – low-grade afternoon fever.
02. **PNEUMONIA** – rusty sputum or pink frothy sputum.
03. **ASTHMA** – **wheezing** on expiration.
04. **EMPHYSEMA** – **barrel chest**.
05. **KAWASAKI SYNDROME** – **strawberry tongue**.
06. **PERNICIOUS ANEMIA** – **red beefy tongue** (need vitamin B12).
07. **DOWN SYNDROME** – **protruding tongue**.
08. **CHOLERA** – **rice watery stool**.
09. **MALARIA** – **step ladder like fever with chills**.
10. **TYPHOID** – **rose spots in abdomen**.
11. **DIPHTHERIA** (infection of nose and throat)– **pseudo membrane formation**
12. **MEASLES** – Koplick's spots are red spots with blue center, usually in the mouth
13. Systemic Lupus E – butterfly rashes.
14. **LIVER CIRRHOSIS** – **spider like varices**.
15. **LEPROSY** – **leonine face** (skin lesions and nerve damage)
16. **BULIMIA** – **chipmunk face**.
17. **APPENDICITIS** – rebound tenderness.
18. **DENGUE** (fever from mosquitos) - **petechiae** or (+) **Herman's sign**.
19. **MENINGITIS** – **Kernig's sign** (leg flex then leg pain on extension), **Brudzinski sign** (neck flex = lower leg flex).

20. **TETANY** – hypocalcemia (+) Trousseau's sign/carpopedal spasm; Chvostek sign (facial spasm).
21. **TETANUS** – **risus sardonicus** (grinning facial spasm).
22. **PANCREATITIS** – **Cullen's sign** (ecchymosis of umbilicus); (+) **Grey turner's spots**.
23. **PYLORIC STENOSIS** – **olive like mass**.
24. **PDA** – **machine like murmur**.
25. **ADDISON'S DISEASE** – bronze like skin pigmentation.
26. **CUSHING'S SYNDROME** – moon face appearance and buffalo hump.
27. **HYPERTHYROIDISM/GRAVE'S DISEASE** – exophthalmos.
28. **INTUSSUSCEPTION** – **sausage shaped mass**, **Dance Sign** (empty portion of RLQ)
29. **Multiple Sclerosis – Charcot's Triad (IAN)**- nystagmus, intention tremor, and scanning or staccato speech.
30. **Multi-Gravitis** – descending muscle weakness
31. **Guillain Barre Syndrome** – ascending muscle weakness
32. **DVT** – **Homan's Sign**
33. **CHICKEN POX** – **Vesicular Rash** (central to distal) **dew drop on rose petal**
34. **ANGINA** – Crushing stabbing pain relieved by NTG
35. **MI** – Crushing stabbing pain which radiates to left shoulder, neck, arms, unrelieved by NTG
36. **Latent TB (latent-TB)** – inspiratory stridor

37. **TEF (trach-esoph-fist)** – 4Cs' Coughing, Choking, Cyanosis, Continuous Drooling
38. **EPIGLOTTITIS** – 3Ds' Drooling, Dysphonia, Dysphagia
39. **HODGKIN'S DSE/LYMPHOMA** – painless, progressive enlargement of spleen & lymph tissues, Reed Sternberg Cells
40. **INFECTIOUS MONONUCLEOSIS** – Hallmark: sore throat, cervical lymphadenopathy, fever
41. **PARKINSON'S** – Pill-rolling tremors
42. **FIBRIN HYALIN** – Expiratory Grunt
43. **CYSTIC FIBROSIS** – Salty skin
44. **DM** – polyuria, polydipsia, polyphagia
45. **DKA** – Kussmaul's breathing (Deep Rapid RR)
46. **BLADDER CA** – painless hematuria
47. **BPH** – reduced size & force of urine
48. **PEMPHIGUS VULGARIS** – Nikolsky's sign (separation of epidermis caused by rubbing of the skin)
49. **RETINAL DETACHMENT** – Visual Floaters, flashes of light, curtain vision
50. **GLAUCOMA** – Painful vision loss, tunnel/gun barrel/halo vision (Peripheral Vision Loss)
51. **CATARACT** – Painless vision loss, Opacity of the

- lens, blurring of vision
52. **RETINOBLASTOMA** – Cat's eye reflex (grayish discoloration of pupils)
 53. **ACROMEGALY** – Coarse facial feature (too much growth hormone)
 54. **DUCHENNE'S MUSCULAR DYSTROPHY** – Gowers' sign (use of hands to push one's self from the floor)
 55. **GERD** – Barrett's esophagus (erosion of the lower portion of the esophageal mucosa)
 56. **HEPATIC ENCEPHALOPATHY** – Flapping tremors
 57. **HYDROCEPHALUS** – Bossing sign (prominent forehead)
 58. **INCREASE ICP** – HYPERTension BRADYpnea BRADYcardia (Cushing's Triad)
 59. **SHOCK** – HYPOtension TACHYpnea TACHYcardia
 60. **MENIERE'S Disease** – Vertigo, Tinnitus
 61. **CYSTITIS** – burning on urination
 62. **HYPOCALCEMIA** – Chvostek & Trousseau's sign
 63. **ULCERATIVE COLITIS** – recurrent bloody diarrhea
 64. **LYME'S Disease** – Bull's eye rash
 65. Basilar Fracture – Otorrhea
 66. **Orbital Fracture** – Battle signs & Raccoon's Eye

Immunology

Sepsis and anaphylaxis (along with the obvious hemorrhaging) reduce circulating volume by way of increased capillary permeability, which leads to **reduced preload** (volume in the left ventricle at the end of diastole).

Allergies

- **Basophils release histamine** during an allergic response.
- Latex allergies □ Assess for allergies to bananas, apricots, cherries, grapes, kiwis, passion fruit, avocados, chestnuts, tomatoes, peaches
- Prior to a CT scan, assess for allergies

Immunizations

- Ask for allergy to eggs before Flu shot
- Age 4 to 5 yrs child needs DPT/MMR/OPV (OPV = Polio vaccine)
- If kid has cold, can still give immunizations
- **MMR and Varicella immunizations come later**, around 15 months.
- MMR
 - The MMR vaccine is given SQ not IM.
 - Ask for anaphylactic reaction to eggs or neomycin before MMR vaccine
- For HIV kids avoid OPV and Varicella vaccinations (live) but give Pneumococcal and influenza.
 - MMR is avoided only if the kid is severely immunocompromised.
- **Pneumovax 23** gets administered post splenectomy to prevent pneumococcal sepsis.
- kids can get vaccines if they have mild illness (fever <101, cold, ear infection, mild diarrhea) but should be related signs and symptoms, if it is moderate-severe. **ok if they are taking antibiotics but not antivirals!**

Leadership

If one nurse discovers another nurse has made a **mistake** it is always appropriate to **speak to her** before going to management. If the situation persists, **then** take it higher.

Delegation

DO NOT delegate what you can EAT!

E – evaluate

A – assess

T - teach

Rules for Delegation

RN ASSIGNMENT

- Cannot delegate assessment, teaching, or nursing judgement

LPN/LVN ASSIGNMENT

- Assign stable with expected outcomes

UAP ASSIGNMENT

- Delegate standard, unchanging procedures

Five Rights of Delegation

RIGHT TASK – scope of practice, stable client

RIGHT CIRCUMSTANCES – workload

RIGHT PERSON – scope of practice

RIGHT COMMUNICATION – specific task to be performed, expected results, follow-up communication

RIGHT SUPERVISION – clear directions, intervene if necessary

Maternity/Women's Health

	Fetal Heart Rate Pattern		Etiology
V	Variable Decels	C	Cord compression
E	Early Decels	H	Head compression
A	Accels	O	Okay, not a problem
L	Late Decels	P	Placental Insufficiency

A= appearance (color all pink, pink and blue, blue [pale])

P= pulse (>100, < 100, absent)

G= grimace (cough, grimace, no response)

A= activity (flexed, flaccid, limp)

R= respirations (strong cry, weak cry, absent)

APGAR measures

Skin color, HR, Reflexes, Muscle tone, RR;

each section is scored between 0-2 points.

0–3 = Severely Depressed (RESUSCITATE)

4–6 = Moderately Depressed

7 –10 = Excellent (OK)

Fetal alcohol syndrome

-Upturned nose-Thin upper lip

1. ABCs (Airway, Breathing, Circulation)

2. **When in distress DO NOT ASSESS!** Unless 2nd hand information is received.

-Flat nasal bridge

-Small for Gestational Age

Rhogam Factor

- **Given at 28 weeks, 72 hours postpartum, IM.**

- **Only given to Rh NEGATIVE mother.**

- **If indirect Combs' test is positive, don't need to give Rhogam, because the mother already has the antibody**

- **Only administer if coombs' test result is negative**

General Notes

- When a patient comes in and she is in active labor, the nurse's first action is to listen to fetal heart tone/rate
- One way to remember which type of measles [regular measles (rubeola) or German measles (rubella)] is dangerous to **pregnant mothers** ~ **Never get pregnant with a German (rubella)**
- Placental abruptio: bleeding with pain, don't forget to monitor volume status (I&O)
- If a laboring mom's water breaks and she is any **minus station**, must better know there is a risk of **prolapsed cord**. (because baby is inside above ischial spine and can compress the cord)
- For **cord compression**, place the mother in the **TRENDELENBURG position** because this removes pressure of the presenting part off the cord. (If her head is down, the baby is no longer being pulled out of the body by gravity) or **hands and knees position**.
- **If the cord is prolapsed, cover it with sterile saline gauze to prevent drying of the cord and to minimize infection.**
- **For late decelerations, turn the mother to her left side, to allow more blood flow to the placenta** (because late decelerations = placenta deficiency)
- For any kind of bad fetal **heart rate** pattern, you give O2, often by mask
- When doing an epidural anesthesia, **hydration before-hand is a priority** (because causes hypotension).
- **Hypotension and bradypnea / bradycardia are major risks and emergencies.**
- **NEVER check the monitor or a machine as a first action. Always assess the patient first!!**
 - For example: **listen to the fetal heart tones with a stethoscope in NCLEX land.**
- Sometimes it's hard to tell who to check on first, the mother or the baby; it's usually easy to tell the right answer if the mother or baby involves a machine. **If you're not sure who to check first, and one of the choices involves the machine, that's the wrong answer.**
- **If the baby is in a posterior presentation, the sounds are heard at the sides.**
- If the baby is anterior, the sounds are heard closer to midline, between the umbilicus and where you would listen to a posterior presentation (because anterior is in front/close to midline)
- **If the baby is breech, the sounds are high up in the fundus near the umbilicus** (because head is up, feet down first)
- If the baby is vertex, they are a little bit above the symphysis pubis. (vertex= head first, so breath sounds right above pubis bone)
- Best way to warm a newborn: skin to skin contact covered with a blanket on mom.
- **Amniotic fluid is alkaline** and turns nitrazine paper **blue**. **Urine and normal vaginal discharge are acidic and turn it pink. Pink for acid and Blue for Alkaline.**
- **Amniotic fluid yellow with particles = meconium stained**
- Cephalohematoma (caput succedaneum) resolves on its own in a few days. This is the type of edema that **crosses the suture lines**.
- **The biggest concern with cold stress and the newborn is respiratory distress!**
- Glucose Tolerance Test for preggies, a result of 140 or higher needs further evaluation.

Medical (Diagnostic) Signs

Murphy's sign – pain with palpation of gallbladder area seen with cholecystitis

Cullen's sign – ecchymosis in umbilical area, seen with pancreatitis

Turner's sign – flank grayish blue (turn around to see your flanks) pancreatitis

McBurney's Point – RLQ pain indicative of appendicitis

LLQ Pain – diverticulitis, low residue, no seeds, nuts, peas

RLQ Pain – appendicitis, watch for peritonitis

Guthrie Test – Tests for PKU, baby should have eaten source of protein first

Allen's test – Occlude both ulnar and radial artery until hand blanches then release ulnar. If the hand pinks up, ulnar artery is good, and you can carry on with ABG/radial stick as planned. **ABGS must be put on ice and whisked to the lab.**

Trendelenburg Test – A **tests for varicose veins**. Pt is supine, and the leg is flexed at the hip and raised above the level of the heart. The veins will empty due gravity or with assistance from the examiner's hand squeezing blood towards the heart → **If they fill proximally = varicosity.**

Babinski Sign – Assessment for nervous system problems.

Toes **curl** = - sign

| Normal in adults & kids above 2 yrs | Abnormal in kids 2 yrs & under

Toes **fan** = + sign

| Abnormal in adults & kids above 2 yrs | Normal in kids 2 yrs & under

Mental Health

Remember with Psych patients, SAFETY is the #1 Priority

Munchausen Syndrome is a psychiatric disorder that causes an individual to self-inflict injury or illness or to fabricate symptoms of physical or mental illness, in order to receive medical care or hospitalization.

In a variation of the disorder, **Munchausen by proxy (MSBP)**, an individual, typically a mother, intentionally causes or fabricates illness in a child or other person under her care.

Neuroleptic malignant **syndrome** (NMS): ~ rxn to antipsychotic medications

-NMS is like S&M;

-you get hot (hyperpyrexia)

-stiff (increased **muscle tone**)

-sweaty (diaphoresis)

-BP, pulse, and **respirations** go up &

-you start to drool

General Notes

- **Tardive Dyskinesia** – irreversible – **involuntary movements of the tongue, face and extremities**, may happen after prolonged use of antipsychotics
- **Depression** often manifests itself in **somatic** ways, such as psychomotor retardation, GI complaints, and pain.
- **For phobic disorders, use systematic desensitization.**
- **Safety** over Nutrition with a severely **depressed** client.
- **Absence of menstruation** leads to **osteoporosis** in the anorexic.

Musculoskeletal/Neurological

ICP AND SHOCK HAVE OPPOSITE V/S

	Shock	Cushing's Triad (r/t to ICP in Brain)
Blood Pressure	↓	↑ (Widening Pulse Pressure - ↑ Systolic/ ↓ Diastolic)
Pulse	↑	↓
Respirations	↑	↓ (Cheyne-Stokes or Irregular Respirations)

Amyotrophic Lateral Sclerosis (ALS) is a condition in which there is a degeneration of motor neurons in both the upper & lower motor neuron systems.

Autonomic Dysreflexia: potentially life-threatening emergency

- Symptoms: HTN, Bradycardia, Pounding H/A, Severe Nasal Congestion
- Elevate head of bed to 90 degree
- Loosen constrictive clothing

- Assess for bladder distention and bowel impaction (trigger) <TREAT THEM>
- Administer antihypertensive meds (may cause stroke, MI, seizure)

Multiple Sclerosis is a chronic, progressive disease with demyelinating lesions in the CNS which affect the white matter of the brain and spinal cord.

Hyperactive deep tendon reflexes, vision changes, fatigue and **spasticity** are all symptoms of **MS**

Motor S/S: limb weakness, paralysis, slow speech

Sensory S/S: numbness, tingling, tinnitus

Cerebral S/S: nystagmus, ataxia, dysphagia, dysarthria

Myasthenia gravis is caused by a disorder in the transmission of impulses from nerve to muscle cell; **worsens with exercise and improves with rest.**

- Give **neostigmine** to pts w/ MG about 45 min before eating, so it can help w/ **chewing & swallowing.**

Myasthenia Crisis: it's used to confirm the diagnosis; a positive reaction to Tensilon – will improve **symptoms**

Cholinergic Crisis: caused by excessive medication-stop med-giving Tensilon will make it worse

Huntington's Chorea: 50% genetic, autosomal **dominant** disorder

S/S: chorea → writhing, twisting, movements of face, limbs and body

–gait deteriorates to no ambulation

–no cure, just palliative care

Myelogram

- Pre-Op – NPO 4-6hr, assess hx of allergies, the table will be moved to various positions during test, **the following meds are withheld 48hr prior; phenothiazines, CNS depressants, and stimulants**
- Post Op – Neuro checks q2-4 hrs, **water soluble HOB up, oil soluble HOB down**, oral analgesics for h/a, encourage PO fluids, assess for distended bladder, inspect site.

Electroencephalography (EEG)

- Pre-Op – 24-48 hrs prior holds meds (especially tranquilizers and stimulants), no caffeine or cigarettes (i.e. stimulants) for 24 hrs prior, meals not withheld, no sleep the night before, pt may be asked to hyperventilate for 3-4 min and watch a bright flashing light
- Post Op – Assess pt for possible seizures, since they're not at greater risk

Cerebral Angiogram

- Pre-Op – well hydrated, lie flat, site shaved, check pulses marked
- Post Op – keep flat for 12-14hr, check site, check pulses, force fluids.

Meningeal Irritation S/S

- Nuchal rigidity
- Positive Brudzinski + Kernig signs
- PHOTOPHOBIA

General Notes

- Lumbar Puncture Post Op – Neuro assessments q15-30 until stable, pt lays flat for 2-3hr, encourage fluids, oral analgesics for headache, observe dressing
- **CSF in meningitis will have high protein, and low glucose.**
- **Decreased acetylcholine is related to senile dementia.**
- **Level of consciousness is the most important assessment parameter with status epilepticus.**
- Hyper reflexes (upper motor neuron issue “your reflexes are over the top”)
- Absent reflexes (lower motor neuron issue)

Nutrition

Fat Soluble Vitamins are A, D, E, K

General Notes

- Be wary of questions regarding children drinking too much milk i.e. more than 3-4 cups of milk each day. Too much milk intake reduces intake of other essential nutrients, especially iron. Watch for **anemia with milk-aholics**.
- **Vitamin D's** presence is required by the **parathyroid gland**, in order for it to function.
- **If the patient is taking digoxin or K-supplements, avoid salt substitutes because many are potassium based**
- **Potassium Sources:** bananas, potatoes, citrus fruits
- **No milk (as well as fresh fruit or veggies) on neutropenic precautions.**
- **Nondairy sources of calcium include RHUBARB, SARDINES, COLLARD GREENS**
- **Nonfat milk** reduces **reflux** by increasing lower esophageal sphincter pressure
- **Yogurt has live cultures, so do not give to immunosuppressed patients**
- **No phenylalanine** with a kid positive for **PKU** (no meat, no dairy, no aspartame).
- **Acid Ash diet:** cheese, corn, cranberries, plums, prunes, meat, poultry, pastry, bread
- **Alk Ash diet:** milk, veggies, rhubarb, salmon

Ortho

Casts:

- You can petal the rough edges of a plaster cast with tape to avoid **skin irritation**.
- **Itching under cast area- cool air via blow dryer, ice pack for 10- 15 minutes. NEVER use Qtip or anything to scratch area**

Walking Devices

COAL (cane walking):

C – Cane
O – Opposite
A – Affected
L – Leg

The cane always moves before the weaker leg.

When **ascending** stairs w/ a cane:

- 1 – Step up with the **stronger leg first**
- 2 – Move the cane next while bearing weight on the stronger leg
- 3 – Finally, move the weaker leg

When **descending** stairs w/ a cane:

1. Lead with the cane
2. Bring the weaker leg down next
3. Finally, step down with the stronger leg

Mnemonic – “**Up** with the **good** and **down** with **bad**.”

- Remember the phrase “**step up**” when picturing a person going up **stairs** with crutches. The good leg goes up first, followed by the crutches and then the bad leg. The opposite happens going down. The crutches go first, followed by the good leg.
- Place a **wheelchair** parallel to the bed on the **side of weakness**

SIGNS of a Fractured hip

- EXTERNAL ROTATION
- SHORTENING
- ADDUCTION

Paget's Disease S/S

- **Tinnitus**
- bone pain
- enlarged/thick bones

General Notes

- **Never release traction UNLESS you have an order from the MD to do so**
- Osteomyelitis is an infectious bone disease □ get blood cultures & antibiotics, then if necessary □ surgery to drain abscess.
- **Pain is usually the highest priority with RA**
- **Swimming is a great exercise for Arthritis**
- **William's position** - Semi Fowlers with knees flexed (Inc. knee gatch) to relieve lower back pain.
 - With low back aches, bend knees to relieve
- Greenstick fractures, usually seen in kids bone breaks on one side and bends on the other

Patient Positioning

1. **Air/Pulmonary Embolism** (S&S: chest pain, difficulty breathing, tachycardia, pale/cyanotic, sense of impending doom) → **turn pt to left side and lower the head of the bed.**
2. **Woman in Labor w/ Un-reassuring FHR (late decels, decreased variability, fetal bradycardia, etc) → turn on left side (and give O2, stop Pitocin, increase IV fluids)**
3. **Tube Feeding w/ Decreased LOC → position pt on right side (promotes emptying of the stomach) with the HOB elevated (to prevent aspiration)**
4. **During Epidural Puncture → side-lying**
5. **After Lumbar Puncture (and also oil-based Myelogram) → pt lies in flat supine** (to prevent headache and leaking of CSF) for 4 to 12 hrs or 2 to 3 hrs as prescribed. **Dressings must be kept sterile & frequent neuro assessments should be performed**
6. **Pt w/ Heat Stroke → lie flat w/ legs elevated (to get fluid to go to head to correct hypotension)**
7. **During Continuous Bladder Irrigation (CBI) → catheter is taped to thigh so leg should be kept straight. No other positioning restrictions.**
8. **After Myringotomy (surgical incision into the eardrum, to relieve pressure or drain fluid).**
→ **position on side of affected ear after surgery (allows drainage of secretions)**
9. **After Cataract Surgery → pt will sleep on unaffected side with a night shield for 1-4 weeks.**
10. **After Thyroidectomy → low or semi-Fowler's, support head, neck and shoulders.**
11. **Infant w/ Spina Bifida → position prone (on abdomen) so that sac does not rupture**
12. **Buck's Traction (skin traction) → elevate foot of bed for counter-traction**
13. **After Total Hip Replacement → don't sleep on operated side, don't flex hip more than 45- 60 degrees, don't elevate HOB more than 45 degrees. Maintain hip abduction by separating thighs with pillows (use a wedge pillow)**
14. **Prolapsed Cord → knee-chest position or Trendelenburg**

15. **Infant w/ Cleft Lip** → position on back or in infant seat to prevent trauma to suture line. While feeding, hold in upright position.
16. **To Prevent Dumping Syndrome** (post-operative ulcer/stomach surgeries) → eat in reclining position or **Low-fowlers (so food doesn't empty and dump so fast)**, lie down after meals for 20-30 minutes (also restrict fluids during meals, **low cholesterol and fiber diet**, small frequent meals)
17. **Above Knee Amputation** → elevate for first 24 hours on pillow, **position prone daily to provide for hip extension.**
18. **Below Knee Amputation** → foot of bed elevated for first 24 hours; **position prone daily to provide for hip extension.**
19. **Detached Retina** → area of detachment should be in the dependent position
20. **Administration of Enema** → position pt in **left side-lying** (Sim's) with knee flexed
21. **After Supratentorial Surgery** (incision behind hairline) → elevate HOB 30-45 degrees
22. **After Infratentorial Surgery** (incision at nape of neck) → position pt **flat** and lateral on either side.
23. **During Internal Radiation** → **on bedrest** while implant in place
24. **Autonomic Dysreflexia/Hyperreflexia** (S&S: pounding headache, profuse sweating, nasal congestion, goose flesh, bradycardia, **hypertension**) → **place client in sitting position (elevate HOB) first before any other implementation**
25. **Shock** → bedrest with extremities elevated 20 degrees, knees straight, head slightly elevated (modified Trendelenburg)
26. **Head Injury** → elevate HOB 30 degrees to decrease intracranial pressure
27. **Peritoneal Dialysis when Outflow is Inadequate** → turn pt from side to side **BEFORE** checking for kinks in tubing (reposition patient, to see if it affects the flow and output of the catheter!)
28. **During a lumbar puncture** → The patient is positioned in lateral recumbent fetal position
29. **Lung Biopsy** → Position the **patient lying on the side of the bed or with arms raised up on pillows over bedside table**, have the patient hold their breath in mid expiration, chest x-ray done immediately afterwards to check for complication of pneumothorax, sterile dressing applied
30. **Pt w/ GERD** → Patient should be lying prone on their **left side**, with **HOB elevated 30 degrees**
31. **Pt w/ Pancreatitis** → Patient should be place in **Fetal position!!**
32. **After Appendectomy** → Position the patient on the **right** side with legs **flexed**. (Puts pressure where appendix was)
33. **Pt w/ Pneumonia** → Lay the pt **on the affected side** to splint and reduce **pain**. If attempting to reduce congestion, the congested lung goes up.
34. **Infant Position while Asleep** → To prevent SIDS, the infant **lays on their back** while asleep, in a bare crib.
35. **During Paracentesis** → Patient should be **semi-fowlers**, or **upright on the edge of the bed**.
36. **During Thoracentesis** → Patient **position patient with arms on pillow on over bed table or lying on side**

Hemoglobin	
Neonates	18 – 27
3 Months	10.6 – 16.5
3 yrs	9.4 – 15.5
10 yrs	10.7 – 15.5

Pediatrics

Injection Sites

- IM
- **Vastus Lateralis for 6 months infants**
 - **Ventrogluteal for Toddlers above 18 months**
 - **Deltoid & Gluteus Maximus for Children**

Developmental Stages

- 2-3 months: turns head side to side
- 4-5 months: grasps, switch & roll
- 6-7 months: **sit at 6** and waves bye-bye
- 8-9 months: stands **straight at eight**
- 10-11 months: belly to butt (phrase has 10 letters)
- 12-13 months: twelve and **up**, drink from a **cup**
- **Stranger anxiety** is greatest **7 - 9 months**
- **Separation anxiety peaks in toddlerhood**
- If you gave a toddler a **choice** about taking medicine and he says no, you should leave the room and **come back** in five minutes, because to a toddler it is another episode. Next time don't ask.
- School-age kids (**5 and up**) are old enough and should have an **explanation** of what will happen a week before surgery such as tonsillectomy.

Interpersonal model (Sullivan) – Behavior motivated by need to avoid anxiety and satisfy needs

- | | |
|--------------------------------|---|
| 1. Infancy 0–18 months | – others will satisfy needs |
| 2. Childhood >6yrs | – learn to delay need gratification |
| 3. Juvenile 6–9 years | – learn to relate to peers |
| 4. Preadolescence 9–12 yrs | – learns to relate to friends of opposite sex |
| 5. Early adolescence 12–14 yrs | – learn independence and how to relate to opposite sex |
| 6. Late adolescence 14–21 yrs | – develop intimate relationship with person of opposite sex |

A child with a **ventriculoperitoneal shunt** will have a small upper-**abdominal incision**. This is where the shunt is guided into the abdominal cavity and tunneled under the skin up to the ventricles. **Assess for possible abdominal distention, since fluid from the ventricles will be redirected to the peritoneum, as well as signs of increasing intracranial pressure.**

ICP in Infants

- Irritability
- Bulging fontanels
- High-pitched cry

ICP in Toddlers

- Lack of appetite
- Headache

Bed-position after shunt placement is flat, so fluid doesn't reduce too rapidly. If S/S of increasing icp are present, **then raise the head to 15-30 degrees.**

Congenital Cardiac Defects

- These defects result in **hypoxia** which the body attempts to compensate for (influx of immature RBC's)
 - Labs supporting this would display: ☐ **Hematocrit, Hemoglobin (compensating for hypoxia and deoxygenated blood) & RBC count**
- Tetralogy of Fallot;
 - When children w/ **Tetralogy of Fallot experience Tet spells, they're treated with morphine.**
 - Think DROP (child drops to floor or squats) or Remember HOPS or POSH

D – defect, septal

R – right ventricular hypertrophy

O – overriding aorta

P – pulmonary stenosis

H – hypertrophy of right ventricle

O – overriding aorta

P – pulmonary stenosis

S – septal defect (ventricle)

P – pulmonary stenosis

O – overriding aorta

S – septal defect (ventricle)

H – hypertrophy of right ventricle

Transesophageal Fistula (TEF) – esophagus doesn't fully develop (this is a surgical emergency)

The **3 C's of TEF** in the newborn:

- 1) **Choking**
- 2) **Coughing**
- 3) **Cyanosis**

Hirschsprung's Ds. → bile is lower obstruction; no bile is upper obstruction; **ribbon like stools.**

- Diagnosed with rectal biopsy looking for **absence of ganglionic** cells.
- Cardinal sign in infants is **failure to pass meconium**, and later the **classic ribbon-like and foul-smelling stools.**

Cystic Fibrosis

- **Respiratory** problems are the chief concern with **CF**
- Give **diet** low fat, high sodium, fat soluble vitamins ADEK
- **Intussusception** common in kids with **CF**. Obstruction may cause fecal emesis, **currant jelly- like stools (blood and mucus)**. A barium enema may be used to hydrostatically reduce the telescoping. Resolution is obvious, with onset of bowel movements.
- **Treatment: Aerosol bronchodilators, mucolytics, and pancreatic enzymes.**

Group-a strep precedes rheumatic fever.

- **Chorea** is part of this sickness (grimacing, sudden body movements, etc.) and it **embarrasses kids.**
- The child will have **joint pain.**
- **Watch for elevated ant streptolysin O to be elevated.**
- **Treatment is Penicillin!!!! Assess for allergy**

Children w/ HIV

- Parents of **HIV+** should wear gloves for care, not kiss kids on the mouth, and not share eating utensils.
- **Western blot**
 - A **positive Western blot** in a child <18 months (presence of HIV antibodies) indicates only that the **mother is infected.**
- **p24 antigen Test**
 - **Two or more positive p24 antigen tests will confirm HIV in kids <18 months.**
 - The p24 can be used at any age.

Prepubescent Penis

- **Hypospadias:** abnormality in which urethral meatus is located on the ventral (back) surface of the penis anywhere from the corona to the perineum (**remember hypo, low (for lower side or under side)**)
- **Epispadias:** opening of the urethra on the dorsal (front) surface of the penis
- Undescended testis or **cryptorchidism** is a known risk factor for **testicular cancer** later in life.
 - Start teaching boy's testicular **self-exam around 12**, because most cases occur during adolescence.

- After a **hydrocele** repair provide **ice bags and scrotal support**.

There's an association between low-set ears and renal anomalies.

- The kidneys and ears develop around the same time in utero. Hence, the **kidneys & ears similarly shaped**.
- When doing an assessment of a neonate, if the nurse notices low set or asymmetrical ears, there is good reason to investigate renal functioning.

Pyloric Stenosis

- The **first sign** of pyloric stenosis in a baby is **mild vomiting that progresses to projectile vomiting**.
- Later a mass may be palpable, the baby will seem hungry often, and may spit up after feedings.
- **Would expect hematocrit and BUN related to dehydration**

What is an **intraosseous infusion** (think into to the bone = intra – osseous (bone))

- In pediatrics, it's a temporary, life-saving measure, for life-threatening emergencies, when iv access cannot be obtained, an osseous (bone) needle is hand-drilled into a bone (usually the tibia), where crystalloids, colloids, blood products and **drugs can be administered into the marrow**.
- When venous access is achieved it can be discontinued.
- **Isoproterenol (blood pressure support medication)**, a beta agonist, is **contraindicated via intraosseous infusion**.

Children in Traction

- What traction is used in a school-age kid with a femur or tibial fracture with extensive **skin damage**?
 - **Ninety, ninety**. The name refers to the angles of the joints.
 - **A pin is placed in the distal part of the broken bone, and the lower extremity is in a boot** cast. The rest are the normal pulleys and ropes commonly seen with balanced suspension.
- A child **hinder should clear the bed when in Bryant's** traction (also used for femurs and congenial hip for young kids).

General Notes

- **Bottle Rot: Do NOT let the mother/father/grandma put anything but water in a child's bottle during naps/over-night** □ **Juice or milk will rot the child's teeth right out of his head**
- It is essential to **maintain nasal patency with children < 1 yr**, b/c they are obligatory nasal breathers.
- **Kawasaki disease causes a heart problem**, specifically, **coronary artery aneurysms** due to the inflammation of blood vessels.
- Kids with RSV; **no contact lenses or pregnant nurses** in rooms where **ribavirin** is being administered by hood, tent, etc.
- **Neonates with heroin withdrawal are irritable and are poor at suckling**
- If you can remove the white patches from the mouth of a baby it is just formula. If you **can't, its candidiasis**.
- Don't pick cough over **tachycardia** for signs of **CHF** in an infant.
- **When performing CPR on an infant check the brachial pulse**
- **Test children for lead poisoning around 12 months of age**
- **body surface area (BSA)** is considered the **most accurate** method for medication dosing with kids.
- In a **five-year old breath once for every 5** compressions doing CPR
- An ill child regresses in behaviors
- **No aspirin** with kids b/c it is associated with Reye's Syndrome, and also no NSAID's such as ibuprofen. **Give Tylenol**.
- **4-year-old kids cannot interpret TIME**. Need to explain time in relationship to a known **COMMON EVENT** (eg: "**Mom will be back after supper**").
- **Toddlers** need to express autonomy (**independence**)
- **Prolonged hypoxemia** is a likely cause of **cardiac arrest** in a child.
- With **omphalocele** and **gastroschisis** (herniation of abdominal contents) dress with **loose saline dressing** covered with **plastic wrap** and **keep eye on temperature**. Kid can lose heat quickly.
- It is always the correct answer to **report suspected cases of child abuse**.

Reproductive Health

General Notes

- **Diaphragm must stay in place 6 hours after intercourse.** They are also fitted so must be re-fitted if you lose or gain a significant amount of weight.
- **Gonorrhea is a reportable disease**
- **Priapism:** painful erection lasting longer than 6 hrs.

Respiratory

Can't cough=ineffective airway clearance

Incentive Spirometry steps

- 1) Sit upright
- 2) Exhale
- 3) Insert mouthpiece
- 4) Inhale for 3 seconds, and then HOLD for 10 seconds

Pulmonary Embolism

- First sign of **PE** is **sudden chest pain**, followed by dyspnea and tachypnea.
- **When O₂ deprived, as with PE, the body compensates by causing hyperventilation (resp alkalosis).**
 - Should the patient breathe into a paper bag?
 - No. If the PaO₂ is well below 80 they need **oxygen**.
 - Look at all your ABG values. As soon as you see the words PE you should think oxygen first.

Carbon Dioxide Narcosis: a reversible state of CNS depression induced by a drug(s).

- **High potassium is expected with this condition,** (hydrogen floods the cell forcing potassium out).
- **Causes** ☐ **intracranial pressure.**

Asthma

- **Coughing** without other s/s is suggestive of **asthma**.
- If child who is **wheezing, stops wheezing**. It could mean he is worsening.
- **If a pt has intercostal retractions – be concerned**
- **The best exercise for asthmatics is swimming** (slow, long, controlled breathing)

Tuberculosis

- If a **TB** patient is unable/unwilling to **comply** with Tx they may need supervision (direct observation).
- TB is a public health risk.

PPD is positive if area of induration is:

- >5 mm in an immunocompromised patient** - lower parameters because immunocompromised
- >10 mm in a normal patient**
- >15 mm in a patient who lives in an area where TB is very rare.**

A **positive PPD** confirms **infection**, not just exposure. A **sputum test** will confirm **active disease**.

Ventilators

- **Complications of Mechanical Ventilation: Pneumothorax, Ulcers**

HO LD

High alarm – **Obstruction** due to increased secretions, kink, patient coughs, gag, or bites the tubing

Low press alarm – **Disconnection** or leak in ventilator or in pt. airway cuff, pt. stops spontaneous breathing

Thoracentesis:

- Pre-Op – Take vital signs, shave area around needle insertion, move the patient into tripod position over a bed table, while holding a pillow
- During – **Withdraw no more than 1000cc at one time**
- Post Op – Listen for bilateral breath sounds, vital signs, check leakage, sterile dressing.

What could cause **bronchopulmonary dysplasia**? (dysplasia means abnormality or alteration)

- **Mechanical ventilation is the primary cause.** Other causes could be infection, pneumonia, or other conditions that cause inflammation or scarring.
- **Premature newborns with immature lungs are ventilated and over time it damages the lungs.**

General Notes

- **TIDAL VOLUME is 7 – 10ml / kg**
- **For COPD pts REMEMBER: 2L Nasal Cannula (2LNC) or less (hypoxic NOT hypercapnic drive)**
 - **PaO2 of 60ish and SaO2 90% = normal (b/c these pts is chronic CO2 retainers)**
- The first sign of ARDS is **increased respirations**. Later comes dyspnea, retractions, air hunger, cyanosis.
- **Signs of hypoxia: restless, anxious, cyanotic tachycardia, increased resps. (also monitor ABG's)**
- **Crackles** suggest pneumonia, which is likely to be accompanied by **hypoxia**, which would manifest itself as mental confusion, etc.
- When using a bronchodilator inhaler conjunction with a glucocorticoid inhaler, administer the bronchodilator first
- **In emphysema the stimulus to breathe is low PO2, not increased PCO2 like normal pts, so don't slam them with oxygen.**
 - Encourage **pursed-lip breathing** which promotes **CO2 elimination**, encourage up to 3000mL/day fluids, high-fowlers and leaning forward.
- **Ambient air (room air) contains 21% oxygen.**
- **Before going for Pulmonary Function Tests (PFT's), a pt's bronchodilators will be with-held and they are not allowed to smoke for 4 hrs prior (so it doesn't alter test results)**
- **Tension pneumothorax trachea shifts to opposite side.**

Transmission-Based Precautions

- **SARS (severe acute resp syndrome) ☐ airborne + contact (just like varicella)**
- **Tetanus, Hepatitis B, HIV are STANDARD precautions**

Airborne

My –	Measles			Measles
Chicken –	Chicken Pox/Varicella	OR	remember MTV	TB
Hez –	Herpes Zoster/Shingles			Varicella
TB--	TB			

- **Disseminated Herpes Zoster is AIRBORNE PRECAUTIONS**

Private Room – **negative pressure with 6-12 air exchanges/hr**, Mask, N95 for TB

Droplet

think of SPIDERMAN!

- S – sepsis
- S – scarlet fever
- S – streptococcal pharyngitis
- P – parvovirus B19

P – pneumonia
P – pertussis
I – influenza
D – diphtheria (pharyngeal)
E - epiglottitis
R – rubella
M – mumps
M – meningitis
M – mycoplasma or meningeal pneumonia
An – Adenovirus

Private Room or cohort Mask, door open, 3 ft distance

Contact

MRS. WEE

M – multidrug resistant organism (MRSA)	V – Varicella zoster
R – respiratory infection	C – Cutaneous diphtheria
S – skin infections ~ VCHIPS	H – Herpes simplex
W – wound infxn	I – Impetigo
E – enteric infxn – clostridium difficile	P – Pediculosis
E – eye infxn – conjunctivitis	S – Scabies

- **Localized Herpes Zoster is CONTACT PRECAUTIONS**
- **A nurse with a localized herpes zoster CAN care for patients as long as the patients are NOT immunosuppressed, and the lesions must be covered!**
- **Hepatitis A is contact precautions**

Triage

Red- Immediate: Injuries are life threatening but survivable with minimal intervention. Ex: hemothorax, tension pneumothorax, unstable chest and abdominal wounds, INCOMPLETE amputations, OPEN fix's of long bones, and 2nd/3rd degree burn with 15%-40% of total body surface, etc.

· Red – unstable, i.e., occluded airway, actively bleeding, see first

Yellow- Delayed: Injuries are significant and require medical care but can wait hrs without threat to life or limb. Ex: Stable abd wounds without evidence of hemorrhage, fix requiring open reduction, debridement, external fixation, most eye and CNS injuries, etc.

· Yellow – stable, can wait up to an hour for treatment, i.e. burns, see second

Green- Minimal: Injuries are minor, and Tx can be delayed to hrs or days. **Individuals in this group should be moved away from the main triage area.** Ex: upper extremity fix, minor burns, sprains, sm. lacerations, behavior disorders.

· Green – stable, can wait even longer to be seen, "walking wounded"

Black- Expectant: Injuries are extensive, and chances of survival are unlikely. Separate but don't abandoned, comfort measures if possible. Ex: Unresponsive, spinal cord injuries, wounds with anatomical organs, 2nd/3rd degree burn with 60% of body surface area, seizures, profound shock with multiple injuries, no pulse, b.p, pupils fixed or dilated.

· Black – unstable clients that will probably not make it, need comfort measures

General Notes

- **DOA – Dead On Arrival**
- Orange tag in triage is non-emergent Psych
- In a disaster you should **triage** the person who is most likely to **not survive last (black code)**

Urinary/Renal System

Peritoneal Dialysis

- It's ok to have abdominal cramps, blood tinged outflow and leaking around site if the Peritoneal Dialysis cath (tenckhoff) was placed in the last 1-2 wks.
- **Cloudy outflow NEVER NORMAL - peritonitis or infection!!**

Kidneys

- **Low magnesium and high creatinine** signal renal failure.
- **Renal impairment:** serum creatinine elevated, and urine clearance decreased
- Glomerulonephritis
 - Take vital signs q4hrs & daily weights
 - **Consider blood pressure to be the most important assessment parameter.**
 - Dietary restrictions you can expect include fluids, protein, sodium, and potassium.
- **WBC shift to the left in a patient with pyelonephritis (neutrophils kick in to fight infection)**
- **Nephrotic syndrome** is characterized by massive **proteinuria** (looks dark and frothy) caused by glomerular damage. **Corticosteroids are the mainstay.** Generalized edema common.
 - Signs: Edema & Hypotension
 - **Turn and reposition (pt is @ risk for impaired skin integrity)**
- A **laxative**/bowel prep is given the night before an **IVP** in order to better visualize the organs.
 - **IVP = Intravenous Pyelogram, an x-ray of the kidneys, ureters, and urinary bladder**
 - Pre-Op – Assess allergies (contrast dye)
- **Kidney Glucose threshold is 180**

General Notes

- **Uremic fetor** □ smell urine on the breath

NCLEX TIPS

1. When getting down to two answers, choose the assessment answer (assess, collect, auscultate, monitor, palpate) over the intervention except in an emergency or distress situation. If one answer has an absolute, discard it. **Give priority to answers that deal directly to the patient's body, not the machines/equipment.**
2. Key words are very important. Avoid answers with absolutes for example: always, never, must, etc.
3. **With lower amputations patient is placed in prone position.**
4. Small frequent feedings are better than larger ones.
5. Assessment, teaching, meds, evaluation, unstable patient cannot be delegated to an Unlicensed Assistive Personnel.
6. LVN/LPN cannot handle blood.
7. Aminoglycosides (like vancomycin) cause nephrotoxicity and ototoxicity.
8. **IV push should go over at least 2 minutes except for adenosine which goes superfast, 2-4 seconds..**
9. If the patient is not a child an answer with family option can be ruled out easily.
10. **In an emergency, patients with greater chance to live are treated first**
11. **ARDS (fluids in alveoli), DIC (disseminated intravascular coagulation) are always secondary to something else (another disease process).**
12. Cardinal sign of ARDS is hypoxemia (low oxygen level in tissues).
13. in pH regulation the 2 organs of concern are lungs/kidneys.
14. edema is in the interstitial space not in the cardiovascular space.
15. **weight is the best indicator of dehydration**
16. **wherever there is sugar (glucose) water follows.**
17. aspirin can cause Reye's syndrome (encephalopathy) when given to children
18. **when aspirin is given once a day it acts as an antiplatelet.**
19. **Use Cold for acute pain (e.g. Sprain ankle) and Heat for chronic (rheumatoid arthritis)**
20. **guided imagery is great for chronic pain.**
21. when patient is in distress, medication administration is rarely a good choice.
22. with pneumonia, fever and chills are usually present. For the elderly confusion is often present.
23. Always check for allergies before administering antibiotics (especially PCN). Make sure culture and sensitivity has been done before administering first dose of antibiotic.
24. Cor pulmonale (s/s fluid overload) is Right sided heart failure caused by pulmonary disease, occurs with bronchitis or emphysema.
25. COPD is chronic, Pneumonia is acute. **Emphysema and Bronchitis are both COPD.**
26. In COPD patients the baroreceptors that detect the CO₂ level are destroyed. Therefore, O₂ level must be low because high O₂ concentration blows the patient's stimulus for breathing.
27. Exacerbation: acute, distress.
28. **Epi always given in TB syringe.**
29. Prednisone toxicity: Cushing's syndrome= buffalo hump, moon face, high glucose, hypertension.
30. 4 options for cancer management: chemo, radiation, surgery, allow to die with dignity.
31. No live vaccines, no fresh fruits, no flowers should be used for neutropenic patients.
32. **Chest tubes are placed in the pleural space.**
33. Angina (low oxygen to heart tissues) = no dead heart tissues. MI = dead heart tissue present.
34. **Mevacor (anti-cholesterol med) must be given with evening meal if it is QD (per day).**
35. Nitroglycerine is administered up to 3 times (every 5 minutes). If chest pain does not stop go to hospital. Do not give when BP is < 90/60.
36. Preload affects amount of blood that goes to the right ventricle. Afterload is the resistance the blood has to overcome when leaving the heart.
37. **Calcium channel blocker affects the afterload.**
38. **For a CABG operation when the great saphenous vein is taken it is turned inside out due to the valves that are inside.**
39. Unstable angina is not relieved by nitroglycerin.
40. Dead tissues cannot have PVC's (premature ventricular contraction). **If left untreated PVC's can lead to VF (ventricular fibrillation).**

41. 1 t (teaspoon)= 5 ml
1 T (tablespoon)= 3 t = 15 ml
1 oz= 30 ml
1 cup= 8 oz
1 quart= 2 pints
1 pint= 2 cups
1 gr (grain)= 60 mg
1 g (gram)= 1000 mg
1 kg= 2.2 lbs.
1 lb.= 16 oz
* To convert Centigrade to F. $F = C + 40$, multiply $9/5$ and subtract 40
* To convert Fahrenheit to C. $C = F + 40$, multiply $5/9$ and subtract 40.
42. **Angiotensin II in the lungs = potent vasodilator. Aldosterone attracts sodium.**
43. REVERSE AGENTS FOR TOXICITY
Heparin= protamine sulfate
Coumadin= vitamin k
Ammonia= lactulose
Acetaminophen= Acetylcysteine
Iron= deferoxamine
Digitoxin, digoxin= Digi bind
Alcohol withdrawal= Librium
- **Methadone is an opioid analgesic used to detoxify/treat pain in narcotic addicts.**
- **Potassium** potentiates dig toxicity.
44. Heparin prevents platelet aggregation.
45. PT/PTT are elevated when patient is on Coumadin
46. Cardiac output decreases with dysrhythmias. Dopamine increases BP.
47. **Med of choice for Ventricular tachycardia is lidocaine**
48. Med of choice for SVT is adenosine or adenocard
49. Med of choice for Asystole (no heartbeat) is Atropine
50. **Med of choice for CHF is Ace inhibitor.**
51. Med of choice for anaphylactic shock is Epinephrine
52. **Med of choice for Status Epilepticus is Valium.**
53. Med of choice for bipolar is lithium.
54. Amiodarone is effective in both ventricular and atrial complications.
55. **S3 sound is normal in CHF, not normal in MI.**
56. **Give Carafate (GI med) before meals to coat stomach**
57. **Protonix is given prophylactically to prevent stress ulcers.**
58. **After endoscopy check gag reflex.**
59. **TPN (total parenteral nutrition) given in subclavian line.**
60. **Low residue diet means low fiber**
61. Diverticulitis (inflammation of the diverticulum in the colon) pain is around LL quadrant.
62. **Appendicitis (inflammation of the appendix) pain is in RL quadrant with rebound tenderness.**
63. **Portal hypotension + albuminemia= Ascites.**
64. Beta cells of pancreas produce insulin
65. **Morphine is contraindicated in Pancreatitis.** It causes spasm of the Sphincter of Oddi. Therefore, **Demerol should be given.**
66. Trousseau and Chvostek signs observed in hypocalcemia
67. **With chronic pancreatitis, pancreatic enzymes are given with meals, not before or after, given with meal.**
68. Never give K+ in IV push.
69. **Mineralocorticoids are given in Addison's disease.**

70. Diabetic ketoacidosis (DKA)= when body is breaking down fat instead of sugar for energy. Fats leave ketones (acids) that cause pH to decrease.
71. DKA is rare in diabetes mellitus type II because there is enough insulin to prevent breakdown of fats.
72. **Sign of fat embolism is petechiae. Treated with heparin.**
73. **For knee replacement use continuous passive motion machine.**
74. **Give prophylactic antibiotic therapy before any invasive procedure.**
75. **Glaucoma patients lose peripheral vision.** Treated with meds
76. **Cataract= cloudy, blurry vision. Treated by lens removal-surgery**
77. **Co2 causes vasoconstriction.**
78. Most **spinal cord injuries** are at the cervical or lumbar regions
79. Autonomic dysreflexia (life threatening inhibited sympathetic response of nervous system to a noxious stimulus- patients with spinal cord injuries at T-7 or above) is usually caused by a full bladder.
80. Spinal shock occurs **immediately** after spinal injury
81. Multiple sclerosis= myelin sheath destruction, disruption in nerve impulse conduction.
82. **Myasthenia gravis= decrease in receptor sites for acetylcholine. Since smallest concentration of ACH receptors are in cranial nerves, expect fatigue and weakness in eye, mastication, pharyngeal muscles.**
83. **Tensilon test given if muscle is tense in myasthenia gravis.**
84. **Guillain-Barre syndrome= ascending paralysis. Keep eye on respiratory system.**
85. Parkinsons = RAT: rigidity, akinesia (loss of muscle movt), tremors. Treat with levodopa.
86. **TIA (transient ischemic attack) mini stroke with no dead brain tissue**
87. CVA (cerebrovascular accident) is with dead brain tissue.
88. **Hodgkin's disease= cancer of lymph is very curable in early stage.**
89. Rule of NINES for burns
- Head and Neck= 9%
 - Each upper ext= 9%
 - Each lower ext= 18%
 - Front trunk= 18%
 - Back trunk= 18%
 - Genitalia= 1%
90. **Birth weight doubles by 6 month and triple by 1 year of age.**
91. **If HR is <100 do not give dig to children.**
92. **First sign of cystic fibrosis may be meconium ileus at birth. Baby is inconsolable, do not eat, not passing meconium.**
93. Heart defects. Remember for cyanotic -3T's(ToF, Truncus arteriosus, Transposition of the great vessels). Prevent blood from going to heart. If problem does not fix or cannot be corrected surgically, CHF will occur following by death.
94. **With right side cardiac cath=look for valve problems (tricuspid valve problems)**
95. With left side in adults look for coronary complications.
96. **Rheumatic fever can lead to cardiac valves malfunctions.**
97. **Cerebral palsy = poor muscle control due to birth injuries and/or decrease oxygen to brain tissues.**
98. ICP (intracranial pressure) should be <2. measure head circumference.
99. **Dilantin level (10-20). Can cause gingival hyperplasia**
100. **for Meningitis check for Kernig's/ Brudzinski's signs.**
101. **Wilm's tumor is usually encapsulated above the kidneys causing flank pain (don't palpate abdomen)**
102. **Hemophilia is x-linked. Mother passes disease to son.**
103. **When phenylalanine increases, brain problems occur.**
104. Buck's traction= knee immobility
105. **Russell traction= femur or lower leg**
106. **Dunlap traction= skeletal or skin**
107. **Bryant's traction= children <3y, <35 lbs. with femur fx.**
108. **Place apparatus first then place the weight when putting traction**
109. **Placenta should be in upper part of uterus**
110. **Eclampsia is seizure. (hypocalcemia) - give calcium**

111. A patient with a vertical c-section surgery will more likely have another c-section.
112. Perform amniocentesis **before 20 weeks' gestation** to check for cardiac and pulmonary abnormalities.
113. **Rh - mothers receive RhoGAM to protect next baby.**
114. anterior fontanelle closes by 18 months. Posterior 6 to 8 weeks.
115. **Caput succedaneum= diffuse edema of the fetal scalp that crosses the suture lines. Swelling reabsorbs within 1 to 3 days.**
116. **Pathological jaundice= occurs before 24hrs and lasts 7 days. Physiological jaundice occurs after 24 hours.**
117. **Placenta previa = there is no pain, there is bleeding. Placenta abruption = pain, bleeding.**
118. **Betamethasone (celestone)=surfactant. Med for lung expansion.**
119. Dystocia= baby cannot make it down to canal
120. Pitocin med used for uterine stimulation
121. **Magnesium sulfate (used to halt preterm labor)** is contraindicated if deep tendon reflexes are ineffective. **If patient experiences seizure during magnesium adm. Get the baby out stat (emergency).**
122. Do not use why or I understand statement when dealing with patients
123. **Milieu therapy= taking care of patient/environment**
124. cognitive therapy= counseling
125. **Crisis intervention=short term.**
126. FIVE INTERVENTIONS FOR PSYCH PATIENTS
- Safety
 - Setting limits
 - Establish trusting relationship
 - Meds
 - Least restrictive methods/environment.
127. **SSRI's (antidepressants) take about 3 weeks to work.**
128. Obsession is to think. Compulsion is to action
129. **If patients have hallucinations redirect them. In delusions distract them.**
130. **Thorazine, Haldol (antipsychotic) can lead to EPS (extrapyramidal side effects)**
131. Alzheimer's disease is a chronic, progressive, degenerative cognitive disorder that accounts for more than 60% of all dementia
132. **Change in color** is always a LATE sign!
133. Let's say every answer in front of you is an **abnormal** value. If **potassium** is there, you can bet it is a problem they want you to identify, because values outside of normal can be **life threatening**. Normal potassium is 3.5-5.0. Even a bun of 50 doesn't override a potassium of 3.0 in a renal patient's priority.
134. Look carefully when you have no idea. In a word like rhabdomyosarcoma you can easily ascertain it has something to do with muscle (myo) cancer (sarcoma). The same thing goes for drug names. For example, if it ends in -ide it's probably a diuretic, as in Furosemide, and Amiloride.
135. **When choosing an answer, think in this manner...if you can only do ONLY one thing to help this patient what would it be? Pick the most important intervention.**
136. **An answer that delays care or treatment is ALWAYS wrong.**
137. **If two of the answers are the exact opposite, like bradycardia or tachycardia → one is probably the answer.**
138. If two or three answers are similar or are alike, none is correct.
139. When asking patients questions NEVER use "why" questions. Eliminate all **"why?"** answer options.
140. If you have never heard of it → please don't pick it!
141. Always deal with actual problems or harm, before potential problems.
142. Always select a "patient focused" answer.
143. An answer option that states "reassess in 15 minutes" is probably wrong.
144. Think positive and you can achieve great things.
145. Think of present and future, the past is gone. Forget your past mistakes and focus on your successes encouraging yourself to greater achievements in the future.
146. Always do your best so you can be proud that you gave it your best shot.
147. Focus on your achievements rather than your failures. If you do find yourself thinking about how you failed, then look at what you managed to do right, and how you could correct what you did for next time.